

Surya Prakash S.K

Linkedin: [suryaprakash-sk](#)

Mobile: +91-9492963477

Email: s.surya1754@gmail.com

RESEARCH SUMMARY

PhD Research Scholar at IIT Mandi's Centre for AI and Robotics, specializing in vision-based real-time single-arm and dual-arm collaborative manipulation, with research spanning kinematics-based motion planning for dual-arm manipulators, dynamics-based manipulator control, RL-driven bi-manual systems, and kinematic-based biped locomotion. My work bridges perception, planning, and control, built and validated within high-fidelity simulation environments using NVIDIA Isaac Sim and Isaac Lab into cohesive robot frameworks designed for real-world deployment.

EDUCATION

- **PhD** IIT Mandi, India
PhD - Robotics and AI; GPA: 8.00 Feb 2022 - Present
Courses: Robot Kinematics and Dynamics, Vision-based Robot Learning, Deep Learning, Control Systems, Pattern recognition, Reinforcement Learning
- **M.Tech** NIT Jamshedpur, India
Master of Technology - Computer Integrated Design and Manufacturing; GPA: 8.60 July 2017 - June 2019
Courses: Advanced Mechanics of Solids, Finite Element Methods, Advanced Robotics
- **B.Tech** AP IIIT, R.K Valley, India
Bachelor of Technology - Mechanical Engineering; GPA: 7.83 July 2011 - June 2015
Courses: Engineering Mechanics, Kinematics and Dynamics of Machinery, Design of Machine Elements, Robotics, Solidworks, Abaqus Simulia

SKILLS SUMMARY

- **Robotics, AI & Vision:** ROS, ROS2, Gazebo, Isaac Sim, Isaac Lab, PyBullet, PyTorch, OpenCV, RGB-D
- **Embedded & Edge Systems:** Arduino, Jetson Nano/Xavier/AGX Orin, Raspberry Pi
- **Tools:** Git, Linux, SolidWorks, Creo, Matlab, ANSYS, L^AT_EX | **Languages:** Python
- **Control & Planning:** Classical Control, Reinforcement Learning, Motion Planning, Trajectory Optimization

EXPERIENCE

- **IIT Mandi** Onsite
Teaching Assistant (Full-time) Feb 22 - Present
 - Robot Kinematics and Dynamics
 - Robot Autonomy
 - Robot Programming Language(ROS1) and Control
- **GVPCE Vishakhapatnam** Onsite
Assistant Professor - Mechanical Engineering Aug 2019 - Feb 2022
 - Supervised Under graduate level final year projects.
 - Handled academic and administrative works in department level.

PROJECTS

- **Improved Artificial Potential Field Method for Dual-Arm Manipulator (Inter-Arm Collision Avoidance, Manipulator, Motion Planning):** Developed an improved Artificial Potential Field (APF) method for dual-arm manipulators to ensure autonomous inter-arm collision avoidance. Implemented a line-segment-based distance matrix for precise proximity monitoring and integrated vision-based pose estimation for target detection. Successfully demonstrated real-time, collision-aware object sorting using the dual-arm system.
Tech: ROS, Deep Learning, OpenCV, Artificial Potential Field, Motion Planning.
- **Competitive Task Allocation with Real-Time Collision Avoidance for Autonomous Dual-Arm Manipulations (Intelligent Task Allocation, spatial-temporal awareness for dual-arm manipulator):** Developed a proximity-based competitive task allocation framework for dual-arm Kinova Gen3 Lite manipulators, integrating YOLOv8-OB vision (30 FPS) with a 10 Hz control loop. Engineered an Improved Artificial Potential Field (iAPF) for real-time inter-arm collision avoidance, utilizing exponential attraction and velocity damping to ensure safety.
Tech: ROS, Intelligent Task Allocation, Deep Learning, OpenCV, Artificial Potential Field, Motion Planning.
- **Continuous Trajectory Tracking for Non-Spherical Wrist Manipulators via Learned Differential Inverse Kinematics (Zero-shot trajectory tracking adaption):** Developed a PPO-based differential IK controller for non-spherical wrist manipulators to solve kinematic coupling and joint-space discontinuities. Trained in Isaac Lab on 1.5M reachable poses, the policy maps a 19-dimensional pose error directly to joint velocities using a custom reward function that balances accuracy with energy efficiency. The system achieved zero-shot sim-to-real transfer on a Kinova Gen3 Lite at 20 Hz.
Tech: Reinforcement Learning, Proximal Policy Optimization.

- **Adaptive Policy Blending in Multi-Agent Reinforcement Learning for Cooperative Dual-Arm Alignment (Dual-Arm Manipulations):** Developed a hierarchical residual MARL framework for coordinated dual-arm manipulation using a Centralized Training and Decentralized Execution (CTDE) paradigm. The system integrates a pretrained PPO base policy with a MAPPO residual policy, utilizing a learned scale-blending parameter (α) to dynamically interpolate between individual arm dexterity and centralized coordination. Trained with a 92-D global critic and 47-D local observations, the framework enables autonomous target picking and pre-assembly alignment, providing a scalable and generalizable alternative to rigid, hard-coded analytical models in industrial automation.
Tech: Reinforcement Learning, Proximal Policy Optimization(PPO), MAPPO.

PUBLICATIONS

- S. P. S. K, A. Shukla, N. Pandya and S. S. Jha, "Automated Vision-based Bolt Sorting by Manipulator for Industrial Applications," 2024 IEEE 20th International Conference on Automation Science and Engineering (CASE), Bari, Italy, 2024, pp. 3602-3607.
- S. P. S. K, A. Shukla and S. S, "Automated Vision-Based Bolt Handling for Industrial Applications Using a Manipulator," 2024 12th International Conference on Control, Mechatronics and Automation (ICCMA), London, United Kingdom, 2024, pp. 193-198.
- Surya Prakash SK, Prajapati D, Narula B and Shukla A (2025) iAPF: an improved artificial potential field framework for asymmetric dual-arm manipulation with real-time inter-arm collision avoidance. *Front. Robot. AI* 12:1604506. doi: 10.3389/frobt.2025.1604506.
- S. P. S. K, D. K. Prajapati, B. Narula, J. Sahoo and A. Shukla, "Kinematics-Based Real-Time Distance Monitoring for Safe Multi-Arm Robot Operations," 2025 25th International Conference on Control, Automation and Systems (ICCAS), Incheon, Korea, Republic of, 2025, pp. 1491-1497.
- Surya Prakash SK et al., "Competitive Task Allocation with Real-Time Collision Avoidance for Autonomous Dual-Arm Manipulations," *Journal of Intelligent & Robotic Systems* (Under Review)
- Surya Prakash SK et al., "ALPHA: Adaptive Learned Policy Hierarchical Blending Architecture for Cooperative Dual-Arm Pre-Assembly Alignment", *IEEE Transactions on Robot Learning* (Under Review)
- Sadhbhav Singh, Surya Prakash SK, Amit Shukla "Prioritized Initial-State Buffers for Agile Drone Racing", *IEEE R-AL* (Under Review)
- Surya Prakash SK et al., "Continuous Trajectory Tracking for Non-Spherical Wrist Manipulators via Learned Differential Inverse Kinematics," *Results in Engineering* (Under review)
Full list of publications are available at [suryaparakash-sk/google scholar](https://scholar.google.com/citations?user=suryaparakash-sk)

REFERENCES

- Dr. Amit Shukla, Associate Professor,
Centre for Artificial Intelligence and Robotics, IIT Mandi,
E-mail : amitshukla@iitmandi.ac.in.
- Dr. Praful Hambarde, Assistant Professor,
Centre for Artificial Intelligence and Robotics, IIT Mandi,
E-mail: praful@iitmandi.ac.in.